Module 2 – Core Service management, Troubleshooting & Professional skills

V2.0

Overview

Day 1

- Service Desk Introduction
- Service Desk Processes

Day 2

- Problem-Solving Framework
- Tools and Techniques
- Case Study: Fast Track Gourmet
- Fishbone Diagram
- Data Gathering
- Review and Implement Solutions
- Documentation and Knowledge Sharing
- Problem-Solving Reflection

How to Complete this Course

This module covers practicals using Miro and GoToMyPC/LOD.

Please ensure you have these installed and working on your own machine.

If you have any issues, please inform your tutor.

How to Complete this Course

As part of your live event, you will have the slides and activity book to complete.

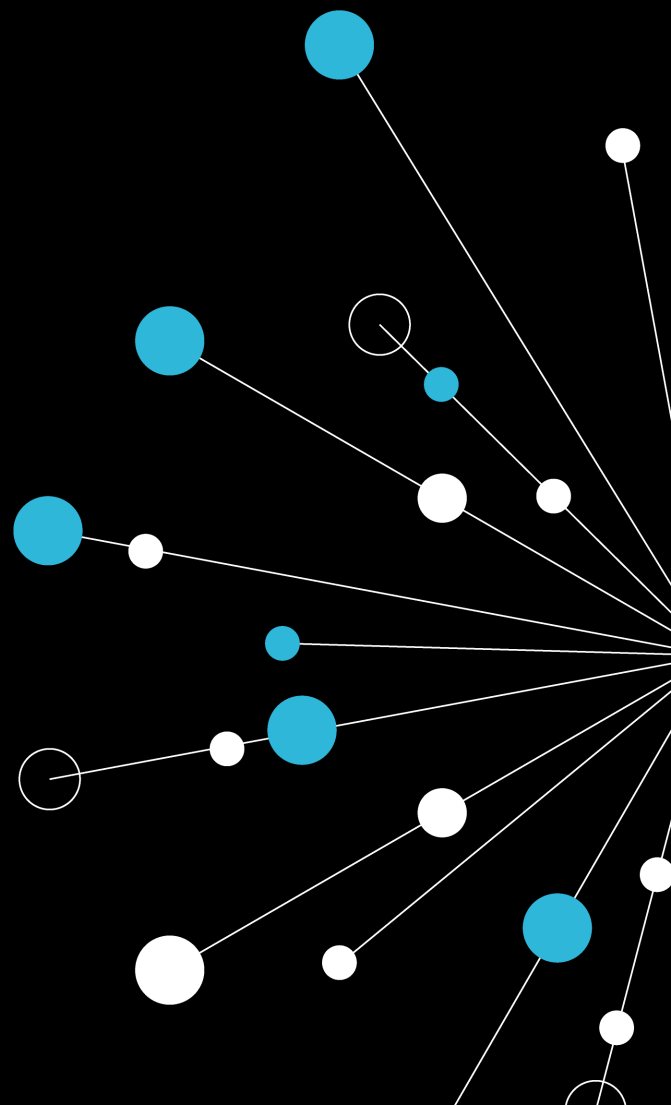
- Open your 'Learner Activity Document' for this course and use it to document your answers.
- Add your full name and date in the 'header' to your Learner Activity Document.
- Use this PowerPoint as a guide to take you through the course.
- Where possible, refer to the business you work for in your answers.
- You may need to ask/do some research to answer some of the criteria.
- Once your Learner Activity Document is completed upload it to Bud along with your Certificate of Completion.
- And most importantly if you need help message or email your tutor.

Introduction

Welcome to this one-day event where you get to put into practice the knowledge and tools you have been introduced to during the Discover part of this module.

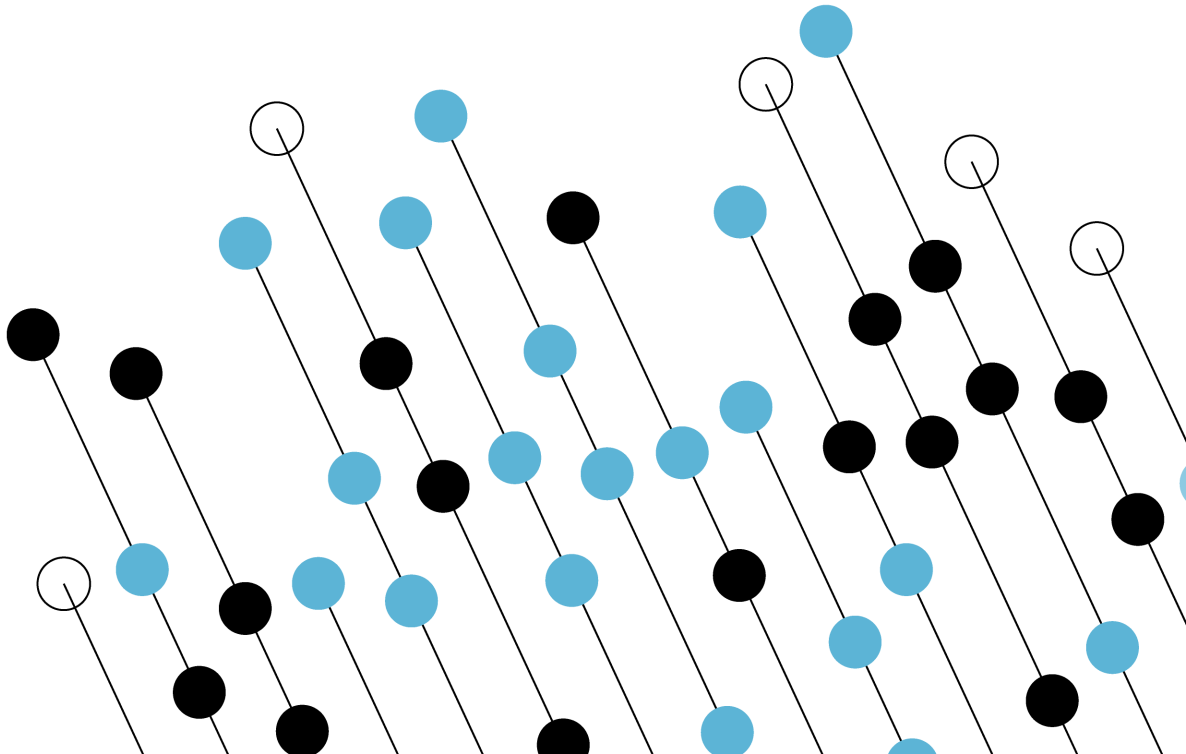
First, let's:

- Introduce ourselves and outline your expectations for the day.
- Familiarise yourself with Miro and how it works.



Service Desk Introduction

Day 1



Service Desk Introduction

In ITIL, a service desk is the central point of contact for users to get help from IT service providers. It's a key part of IT service management, and is responsible for:

- Handling incidents and service requests
- Providing first-line support
- Escalating complex issues
- Managing knowledge
- Providing self-service

The goal of an IT service desk is to provide high-quality service to customers in a timely manner.

Service Desk Introduction

Handling incidents and service requests

The main objective of the service desk is to log and resolve incidents and service requests using a ticket logging system.

Incidents: An *incident* is any unplanned interruption or disruption to an IT service, or a reduction in the quality of that service. It could be anything from a system outage, hardware failure, software malfunction, or network issue that impacts a user's ability to work.

Goal: The primary objective when handling an incident is to restore normal service operation as quickly as possible to minimise disruption. Incident management focuses on resolving the issue and ensuring that the service is up and running again.

Service Desk Introduction

Providing first-line support

In the IT service desk, providing first-line support refers to the initial assistance offered to users or customers who encounter technical issues or require help with IT services.

Basic Troubleshooting and Resolution First-line support addresses common issues such as:

- Password resets
- Software installation help
- Account access issues
- Simple hardware troubleshooting

Support staff use predefined scripts and knowledge bases to resolve these quickly.

Service Desk Introduction

Escalating complex issues

Escalation to Higher Support Levels is a crucial process within IT service desks, particularly for first-line support teams.

When an issue cannot be resolved by the initial support team, it is escalated to ensure that a higher level of expertise or specialised resources are applied to solve the problem efficiently.

Service Desk Introduction

Managing knowledge

Knowledge sharing is a key component of effective IT service desk operations. It involves creating, organising, and maintaining a knowledge base that allows support staff to access information quickly, resolve issues efficiently, and provide consistent support to users.

A well-managed knowledge system benefits both the IT team and the users by reducing response times, minimising repeated issues, and promoting self-service solutions.

Service Desk Introduction

Providing self-service

In an IT service desk environment, providing self-service options empowers users to solve common issues on their own without needing to contact support. Self-service capabilities improve efficiency by reducing the number of tickets raised, allowing the IT team to focus on more complex issues.

A self-service portal serves as a centralised hub where users can log in to access all available resources, from FAQs to request forms. Through the portal, users can:

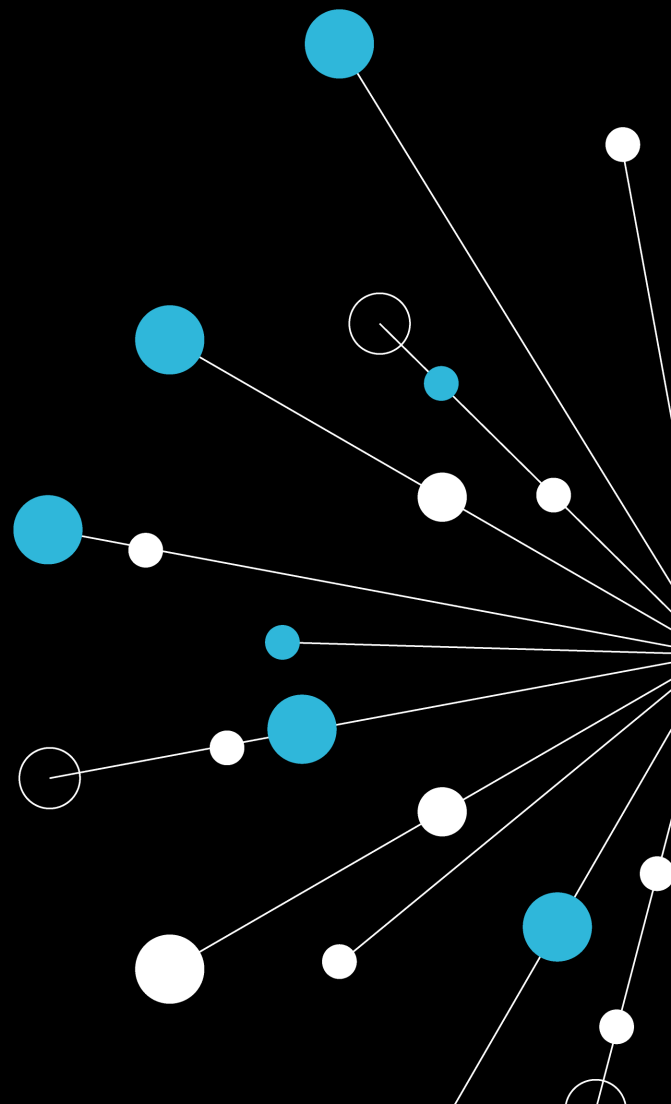
- Search for solutions to common issues
- Submit requests for new software or hardware
- Check the status of existing support tickets
- Access training materials and guidelines for using company software

Service Desk Introduction

Activity 1

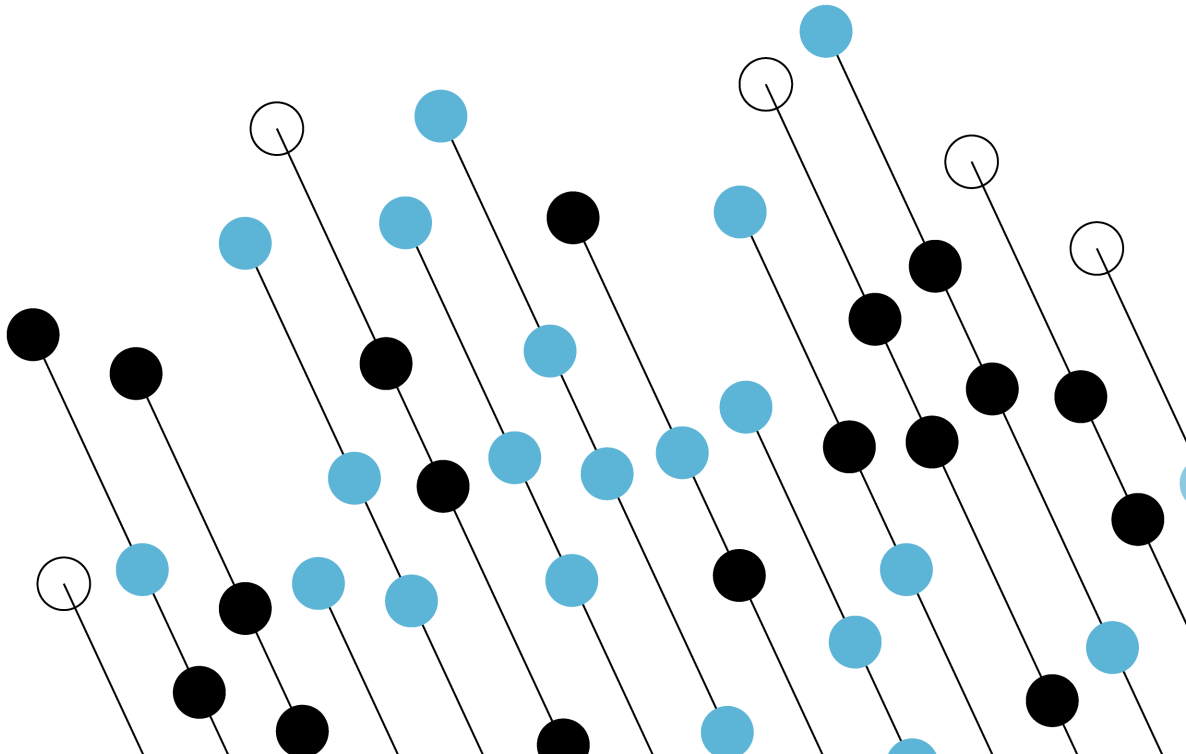
When describing your place of work and processes within the IT service desk, aim to cover the following points:

- Ticket Logging System
- Applications and Services
- Service Level Agreements (SLA)
- Incident Management
- Request Escalations



QA

Service Desk Processes



Service Desk Processes

This topic will delve into different service desk tickets that require resolution. Some of these tickets will necessitate the use of both Packet Tracer and GoToMyPC.

You will:

- Review example user tickets, focusing on triage, escalation steps, and required information for resolution.
- Explore effective communication methods and timing with stakeholders to ensure transparency.
- Investigate tickets involving external customers to illustrate GDPR principles.
- Cover techniques for prioritising workload in a queue.

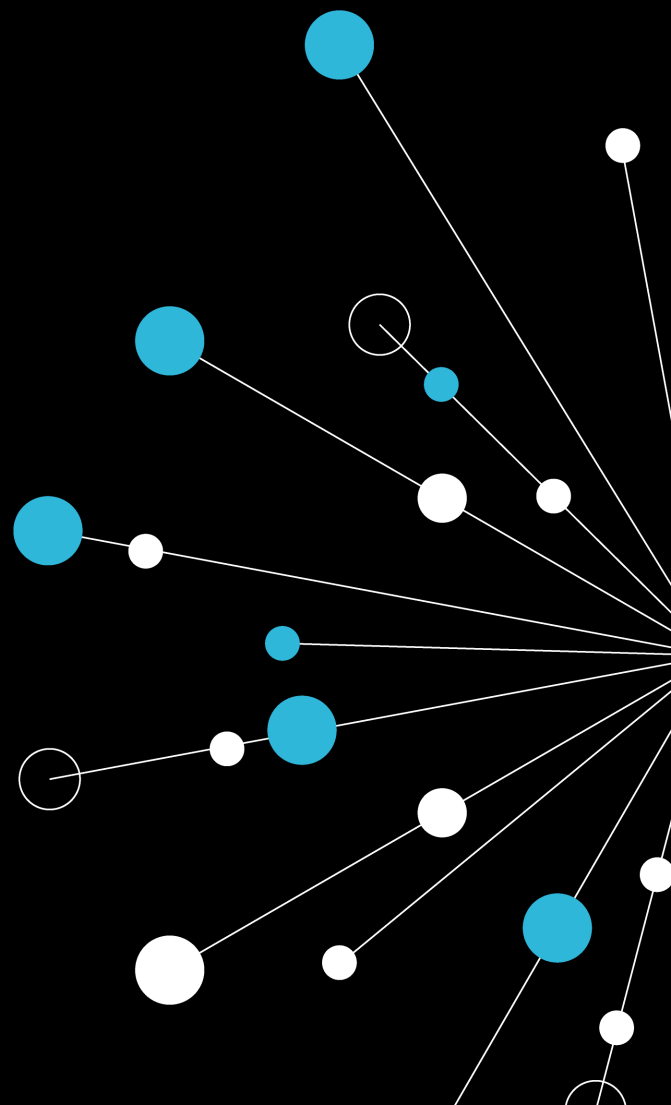
Service Desk Processes

Activity 2

Work through each ticket in the *Learner Activity Document*.

- Resolve the tickets within the practical as described
- Write up your resolution for each ticket

We will discuss the resolution at the end of each ticket.

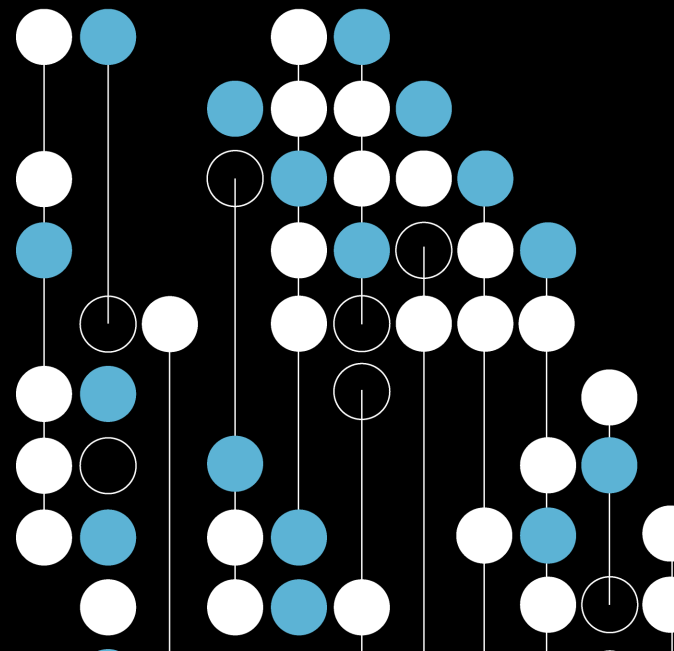


Service Desk Processes

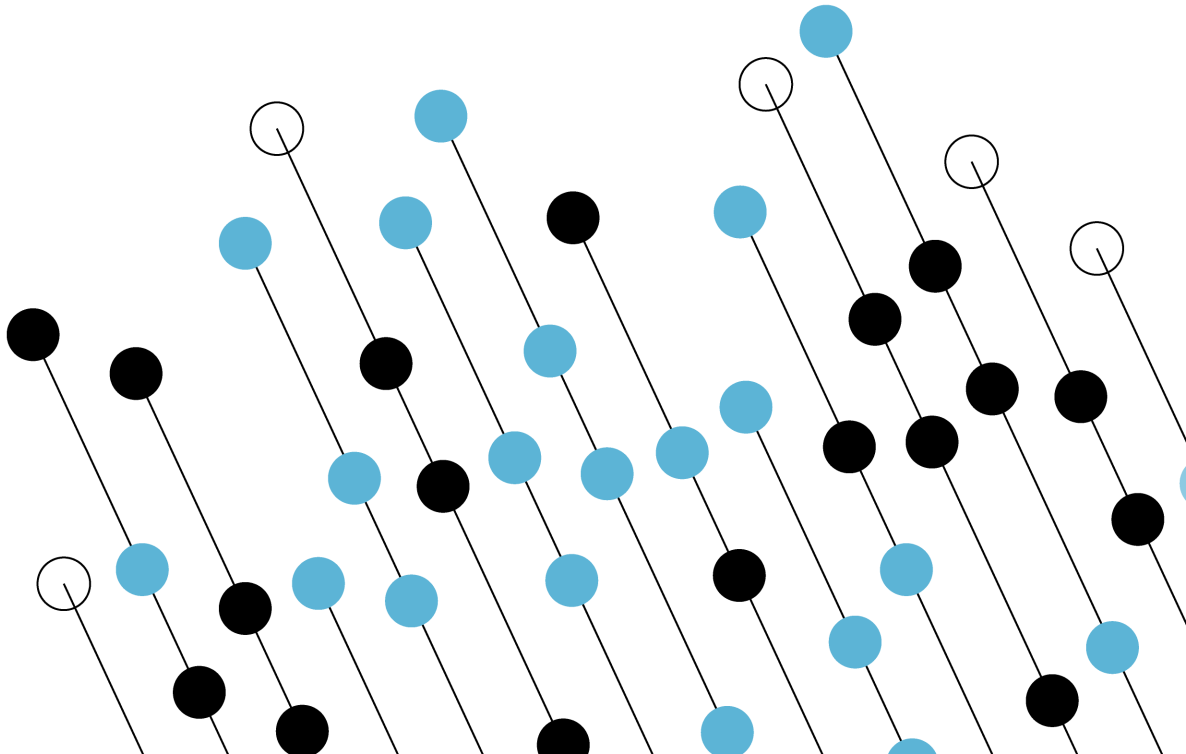
Helpdesk Operator	(yourself)
Ticket Reference Number	1000Example
Customer Name	Evelyn Scott
Date of report	01/02/2024
Description of Issue	"My computer is not recognizing any of my USB devices (keyboard, mouse, external drive). nothing seems to work. Can someone check what's wrong?"
What information must be captured	Screenshot any error messages Check USB on another port on the computer Check USB on another machine Check USB drivers, ensure the hardware is functioning, and update the operating system. Escalate if further diagnostics are needed for hardware issues.
Appropriate methods of communicating	Email communication – to obtain screenshots Telephone to further assist Potentially Remote on the users machine
Description of Resolution	**Check the USB Devices** **Restart the Computer** **Try Different Ports** **Device Manager (Windows)** **System Preferences (Mac)** **Update USB Drivers** **Roll Back Drivers** **BIOS/UEFI Settings** **Enable USB Ports** **Update BIOS/UEFI** **Check user is part of any policies that maybe blocking them**

End of Module Review

- Service Desk Introduction
- Service Desk Processes



Problem-Solving Framework



Problem-Solving Framework

In this section of the course, you will use the materials provided in the Learner Activity Document, along with the content on the Miro Board (link to be provided), to complete assigned tasks and answer questions.

These resources are designed to guide your learning and help you engage with key concepts.

Make sure to carefully review both the document and the Miro Board content, as they contain essential information and examples needed to complete the activities effectively.

Problem-Solving Framework

Let's revisit the framework you were introduced to during the Discover (digital) phase. Looking at the options below, can you put them in the correct step order?

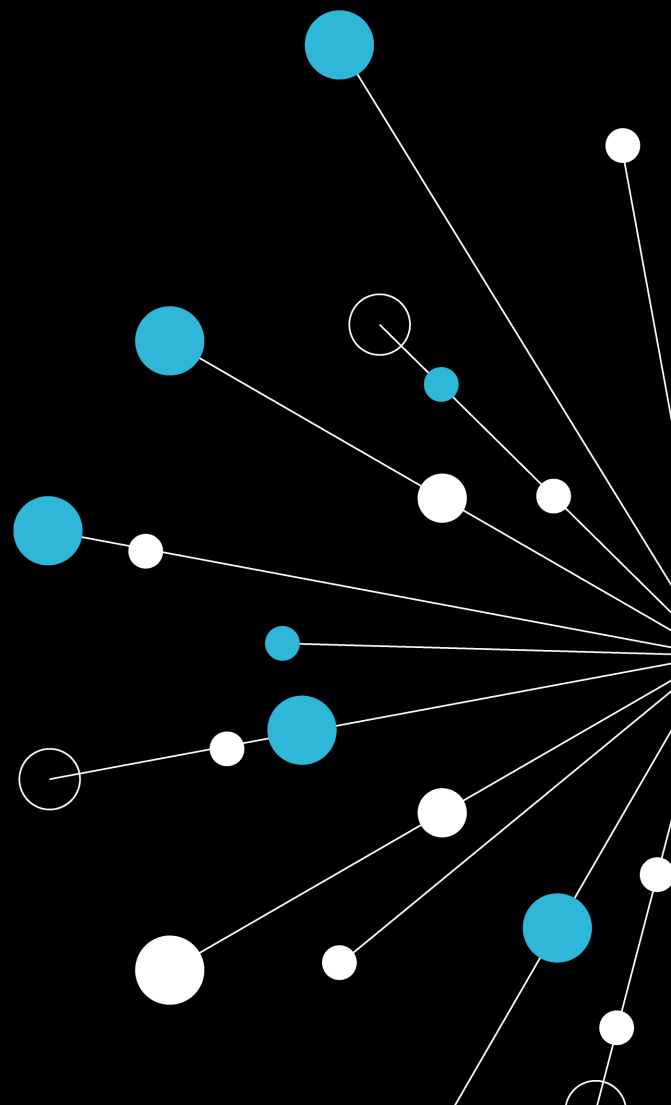
Follow up and review	Select and present solutions for considerations
Implement the solution	Identify the problem
Identify possible solutions	Analyse the problem and gather information

Problem-Solving Framework

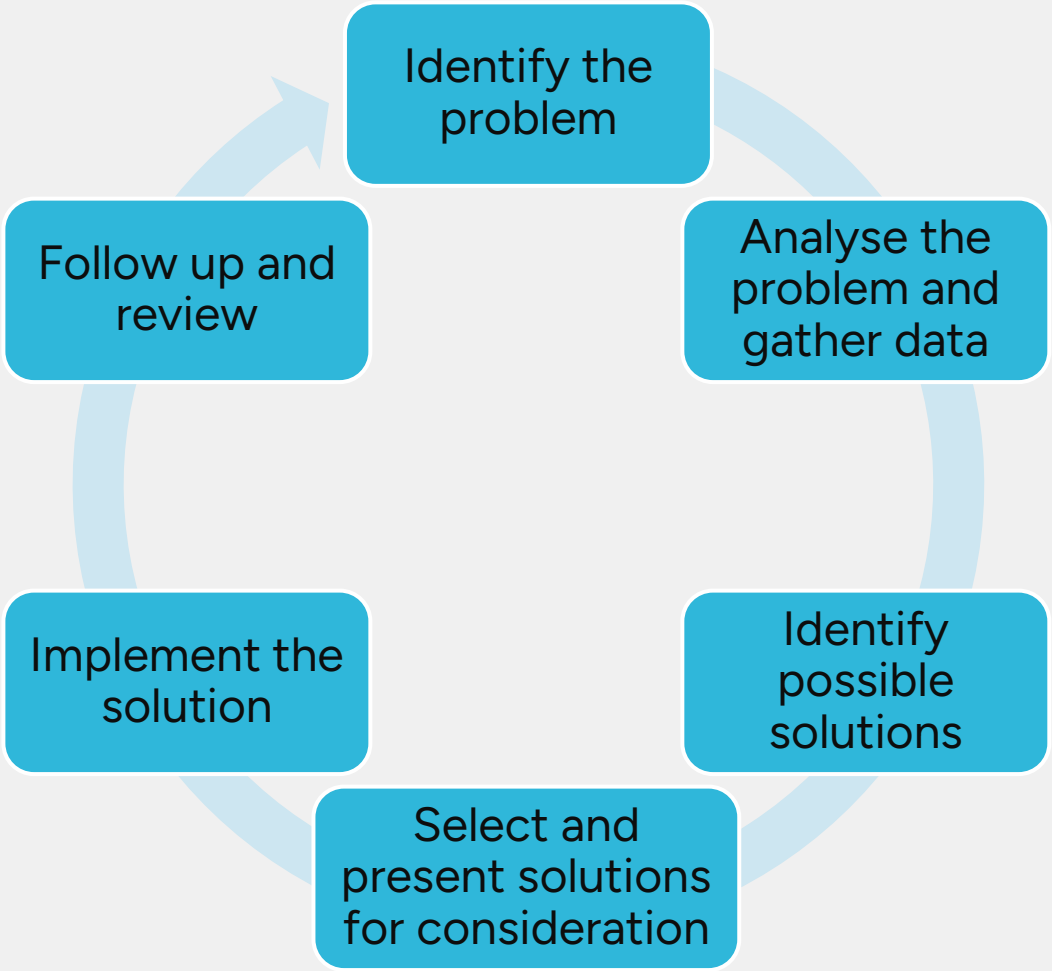
Activity 3

As a group, complete the Problem-Solving Framework activity on Miro.

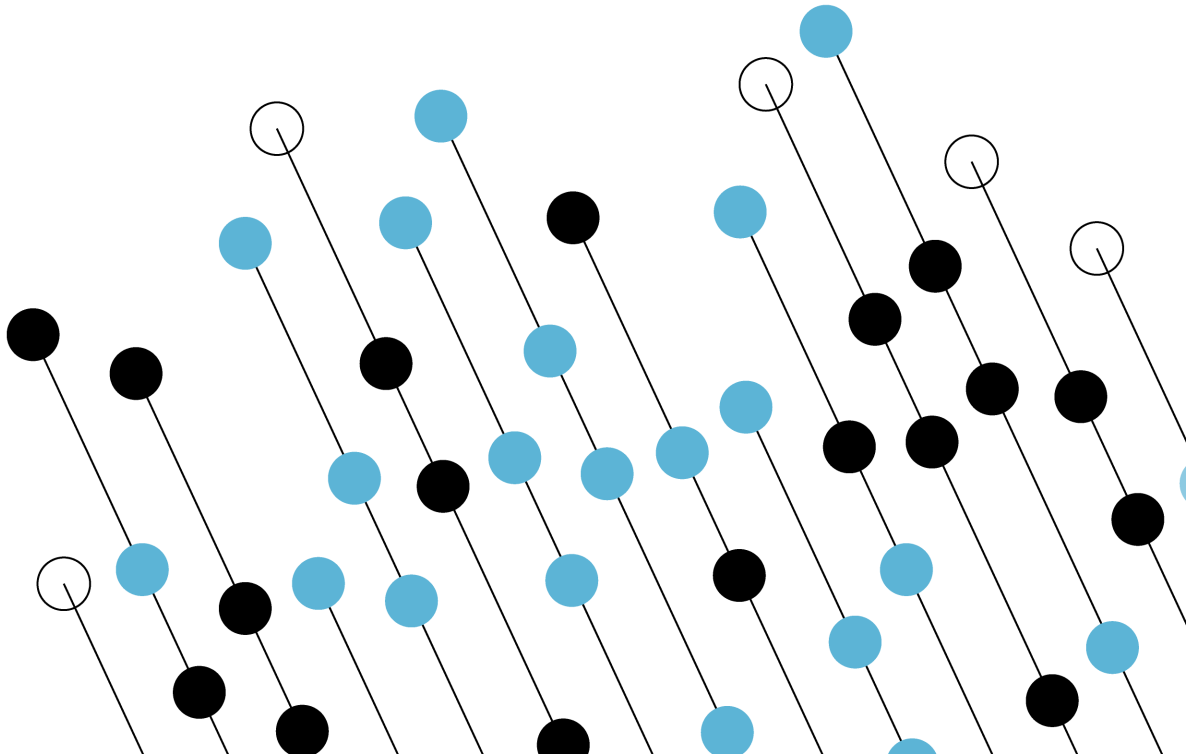
Take a screenshot of your response and include it in your Learner Activity Document.



How did you do?



Tools and Techniques



Tools and Techniques

Which tools and techniques could be used to help solve each of these issues? Some tools could be used in more than one place.

Problem situations	Suggested techniques
Complex problems where a sequence of events needs to be assembled to determine exactly what happened	
Uncertainty over which problems should be addressed first	
Uncertain whether a presented root cause is truly the root cause	
Intermittent problems that appear to come and go and cannot be recreated or repeated in a test environment	
Uncertainty over where to start for problems that appear to have multiple causes	
Struggling to identify the exact point of failure for a problem	
Uncertain where to start when trying to find the root cause	

Tools and Techniques

Here is a list of the problem techniques:

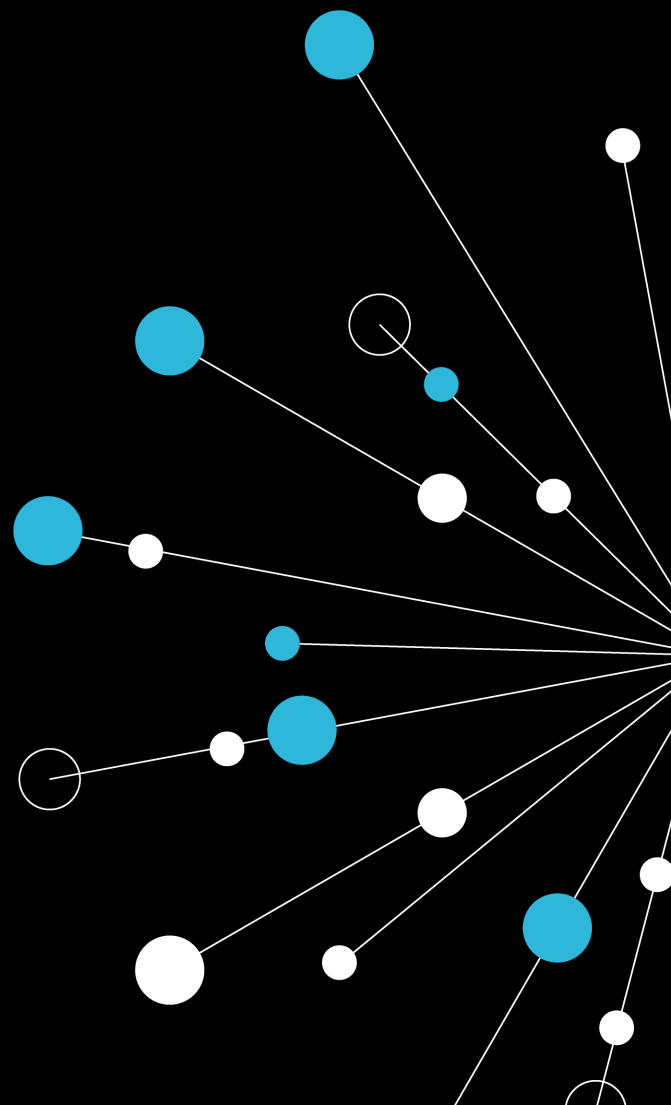
- Pain Value Analysis
- Fault Tree Analysis
- Chronological Analysis
- 5 Whys
- Brainstorming
- Pareto Analysis
- Technical Observation
- Ishikawa / Fishbone

Tools and Techniques

Activity 4

On MIRO match these techniques to the relevant issues detailed on slide 5 (they can be used more than once).

Then in your Learner Activity Document, screenshot your answer and write a summary of the 7 points.

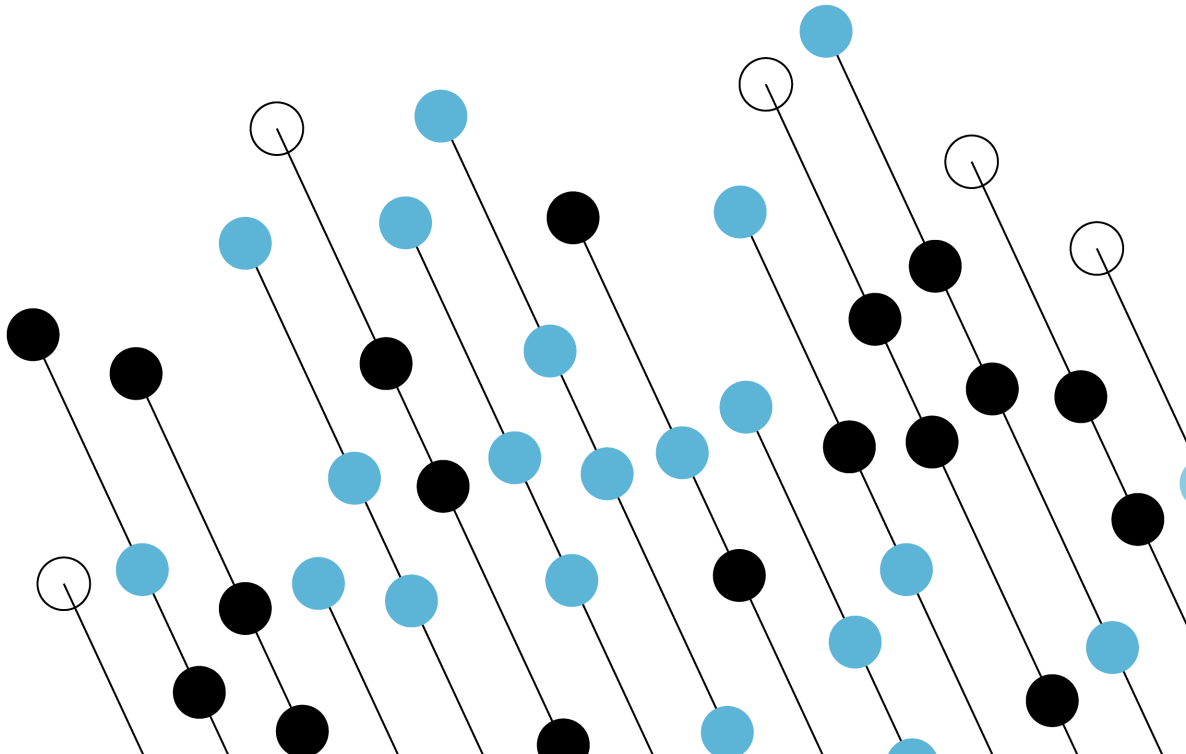


Tools and Techniques

Here is a list of the problem techniques:

Problem situations	Suggested techniques
Complex problems where a sequence of events needs to be assembled to determine exactly what happened	• Chronological analysis • Technical observation post
Uncertainty over which problems should be addressed first	• Pain value analysis • Brainstorming
Uncertain whether a presented root cause is truly the root cause	• 5 whys
Intermittent problems that appear to come and go and cannot be recreated or repeated in a test environment	• Technical observation • Brainstorming
Uncertainty over where to start for problems that appear to have multiple causes	• Pareto analysis • Ishikawa diagrams • Brainstorming
Struggling to identify the exact point of failure for a problem	• Fault isolation • Ishikawa diagrams • Brainstorming
Uncertain where to start when trying to find the root cause	• 5 whys • Brainstorming

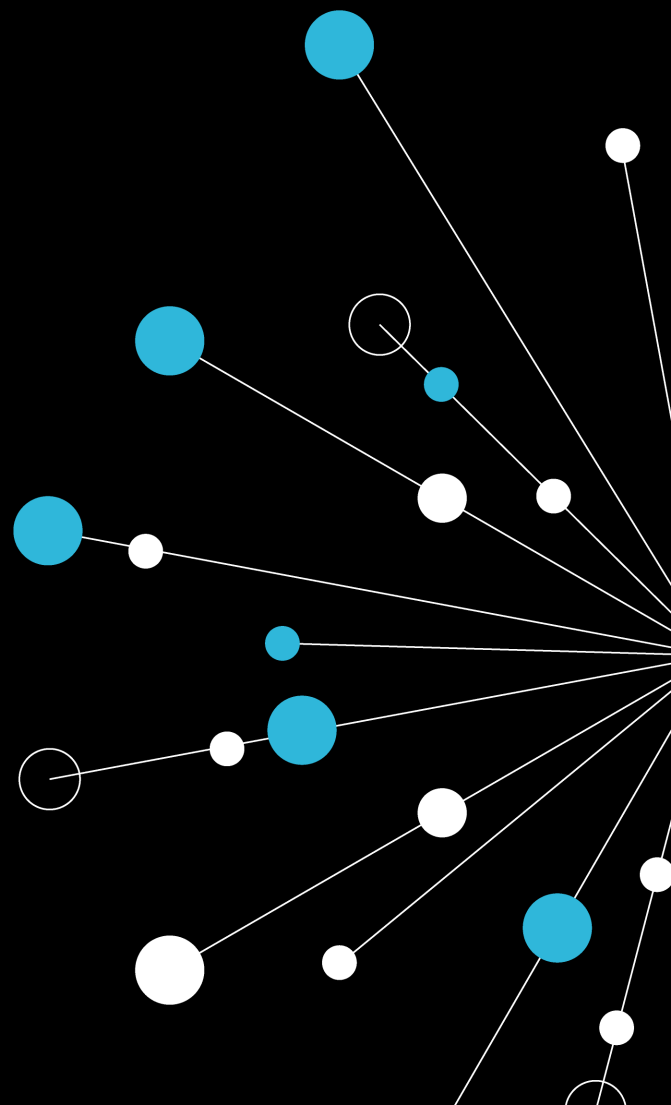
Case Study Assessment



Fast Track Gourmet Case Study

Let's now start putting the tools and skills you worked through within the Discover stage into practice.

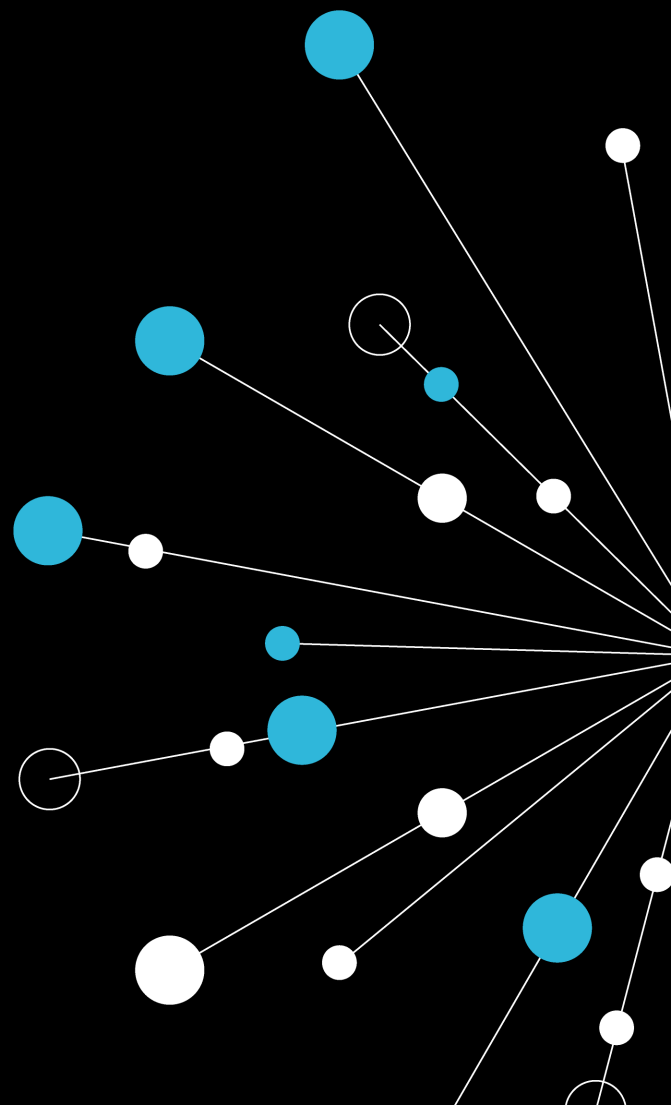
- Read through the case study: Fast Track Gourmet.
- Think about the problems that the company are facing.
- Take notes to summarise the key points.



Fast Track Gourmet Case Study

Return to your groups and complete the following activity using the case study:

- Work through the problems you have identified with your team.
- Agree with your team which three problems need further investigation.



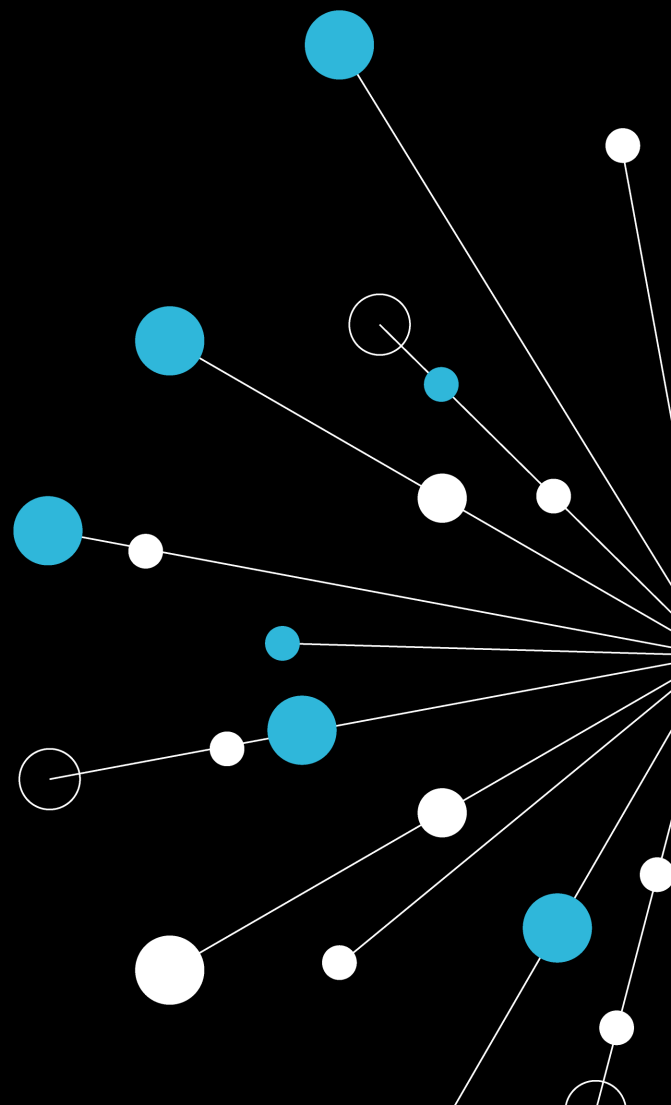
Fast Track Gourmet Case Study

Activity 5

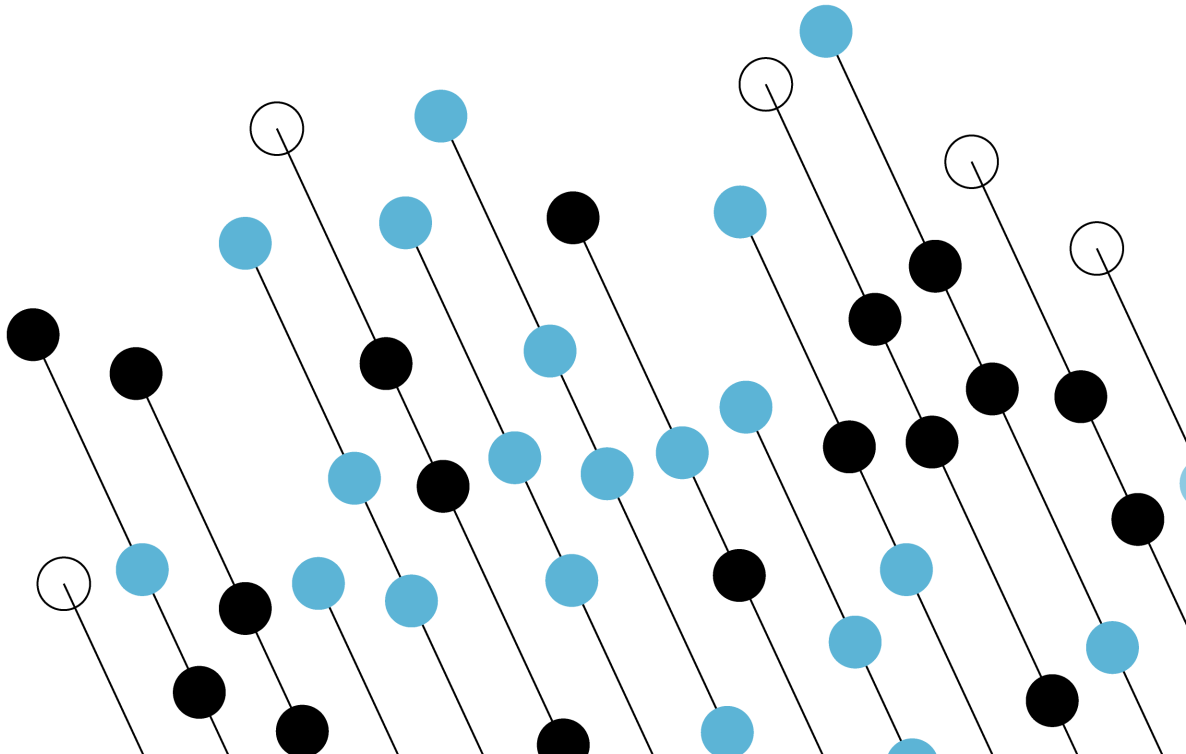
You need to create a clear problem statement for each of the problems you have identified. It needs to describe the problem and its impact in such a way that people are keen to solve it.

It is important to check all assumptions around perceived problems – this will include understanding:

- The cause
- The symptoms
- Any preferred solutions



Fishbone Diagram

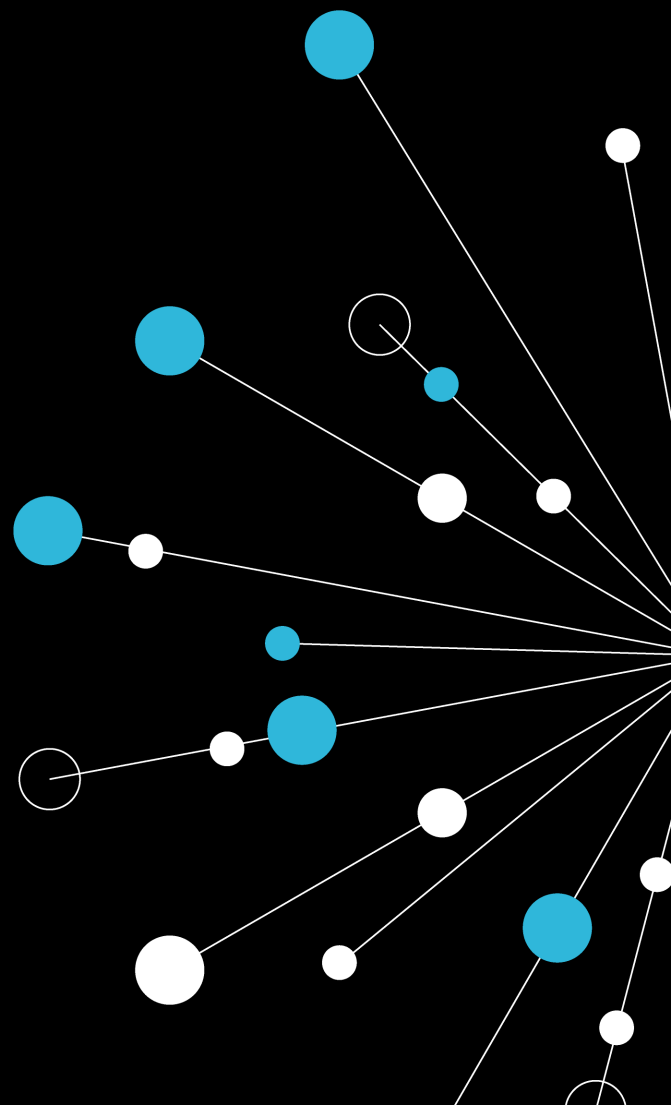


Fishbone Diagram

Activity 6

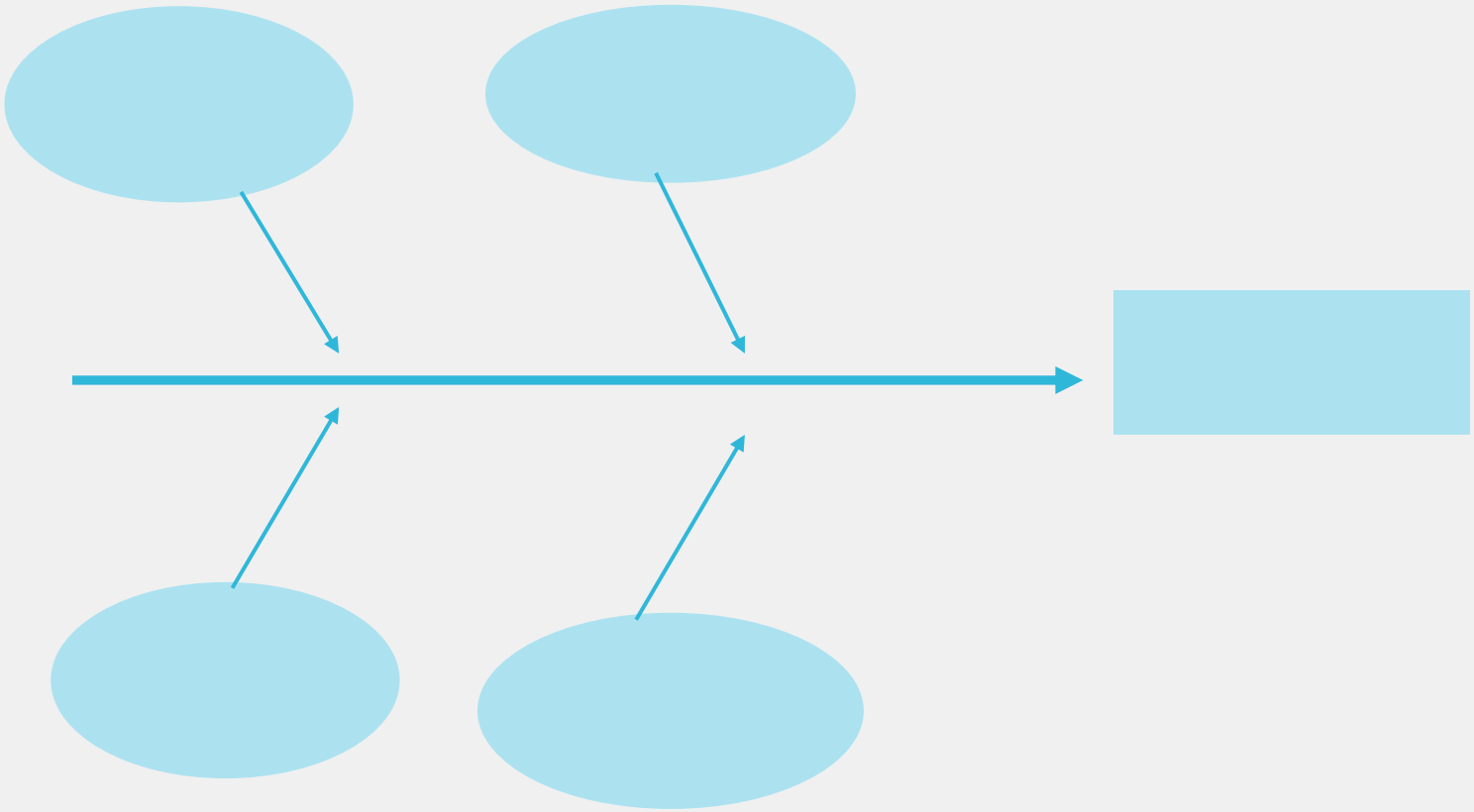
Using one of the problem statements you have defined, create a Fishbone diagram on Miro. You can copy and paste the arrows to build your diagram.

In your Learner Activity Document, screenshot your diagram and write up a short summary.

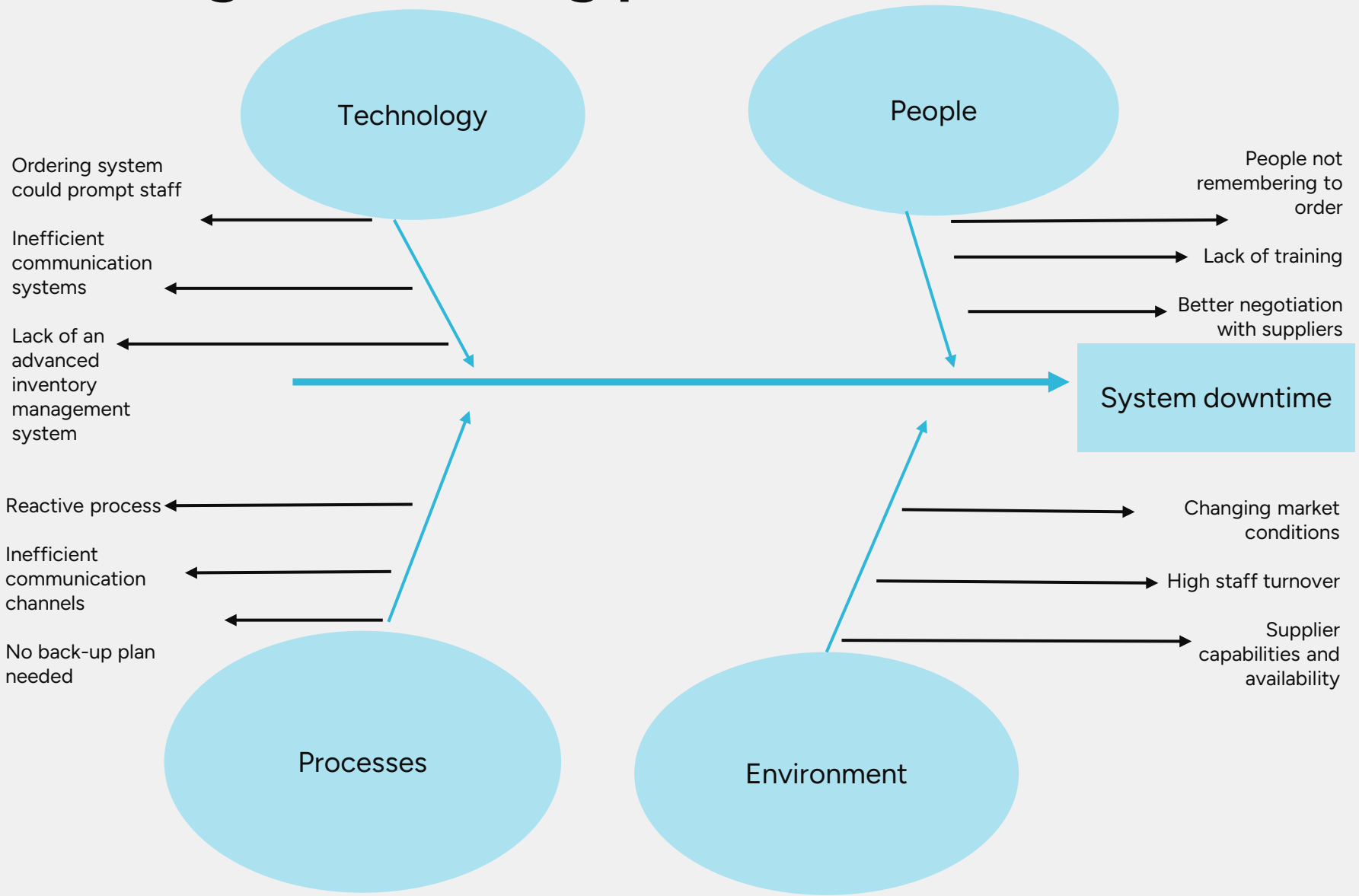


Fishbone Diagram

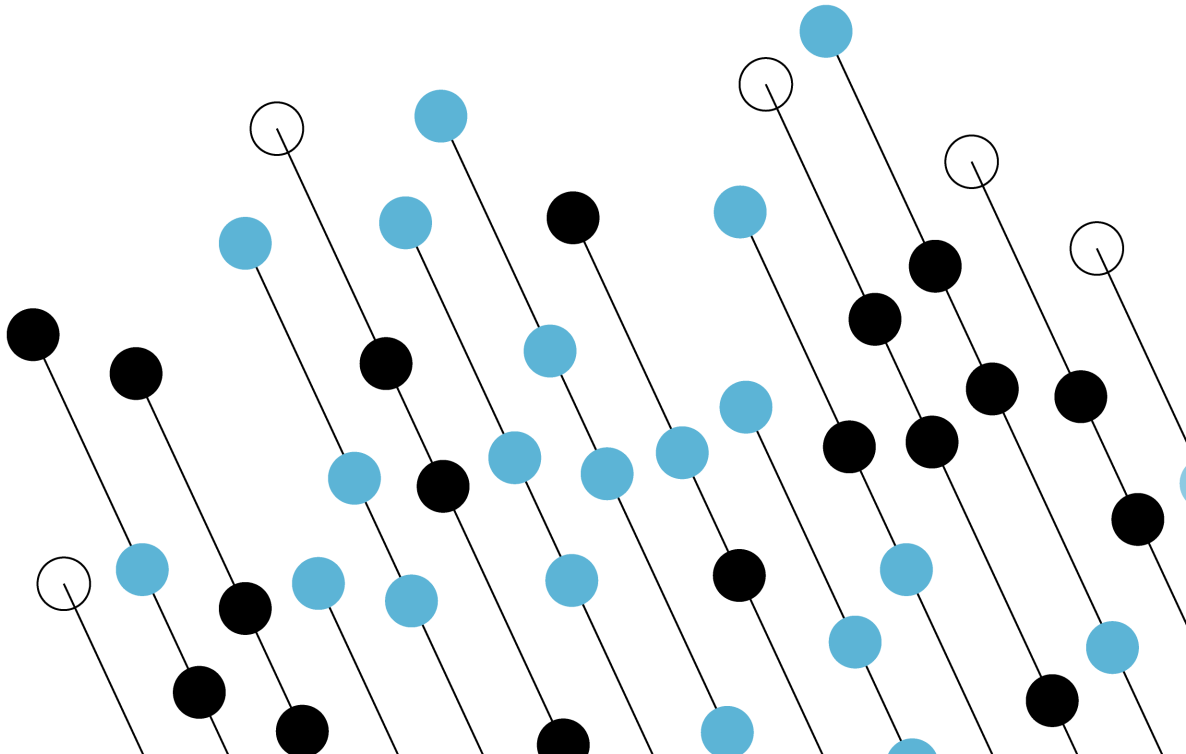
Using one of the problem statements you have defined, create a fishbone diagram.



Fishbone Diagram showing possible causes



Data Gathering



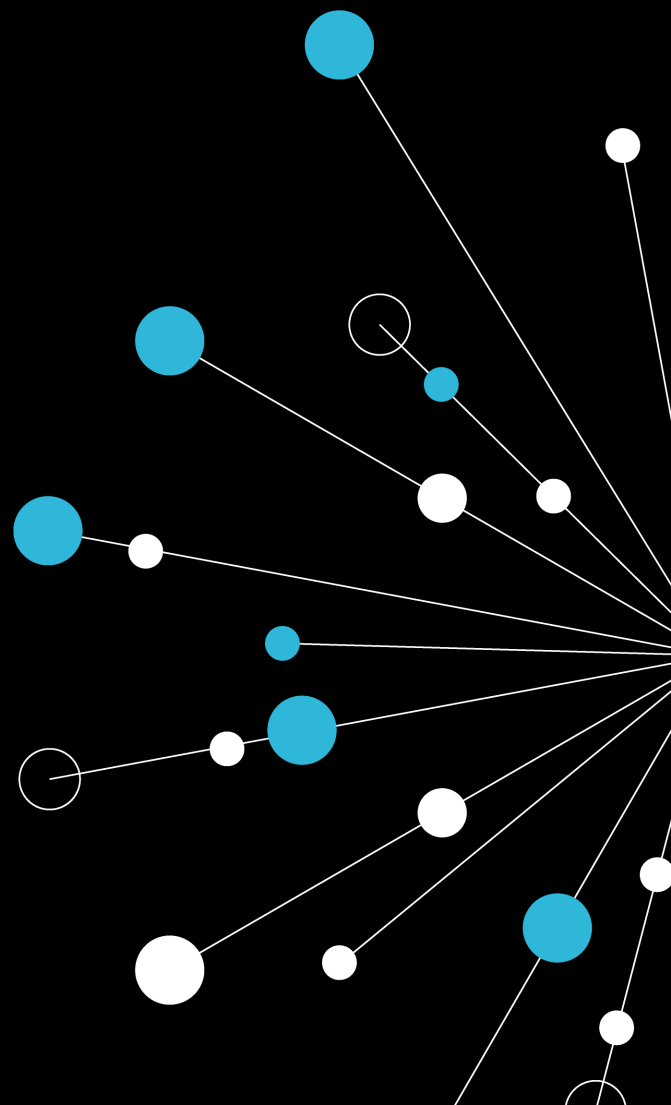
Data Gathering for Fast-Track Gourmet

Activity 7

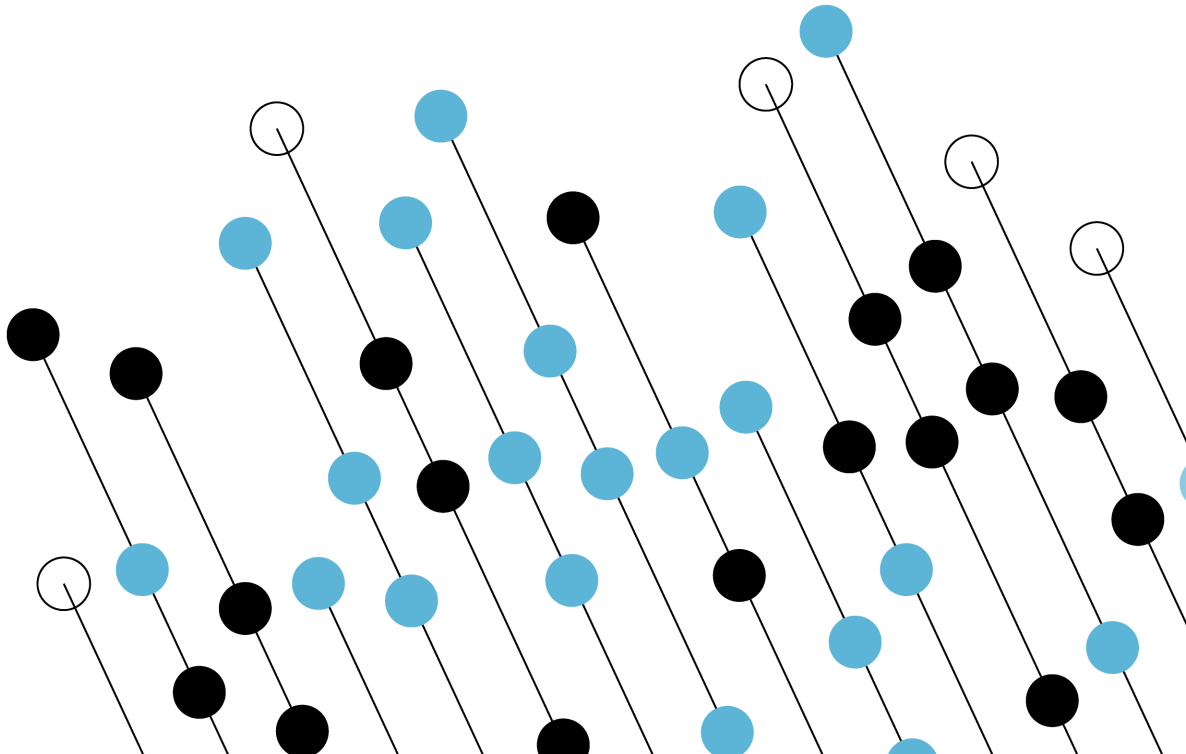
In your group, look back at the three problems you identified earlier and their associated statements.

Brainstorm the data you will need to collect to support your understanding of the three problems and develop a checklist for each problem which specifies the data required and how this will be collected.

Review your lists to check for any duplication of effort across the three problems.



Review and Implement Solutions



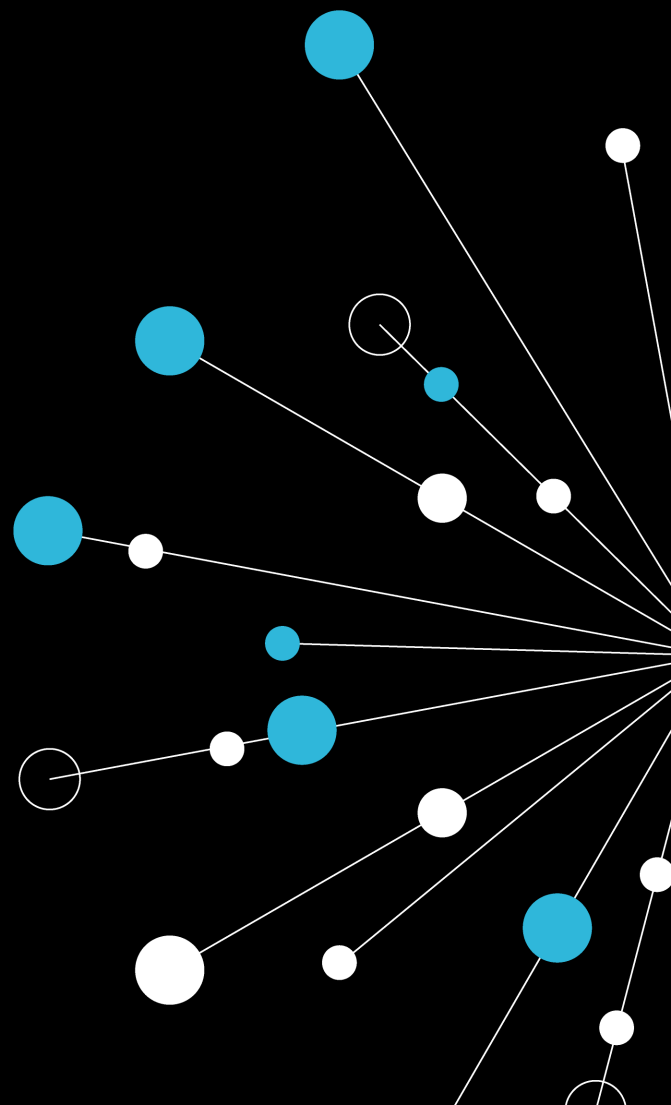
Review so far

Activity 8

Gather information to understand the problem more effectively, select the best solution option from a group of creative ideas.

What still needs to be accomplished?

In your Learner Activity Document, screenshot your diagram from Miro and write up a short summary.

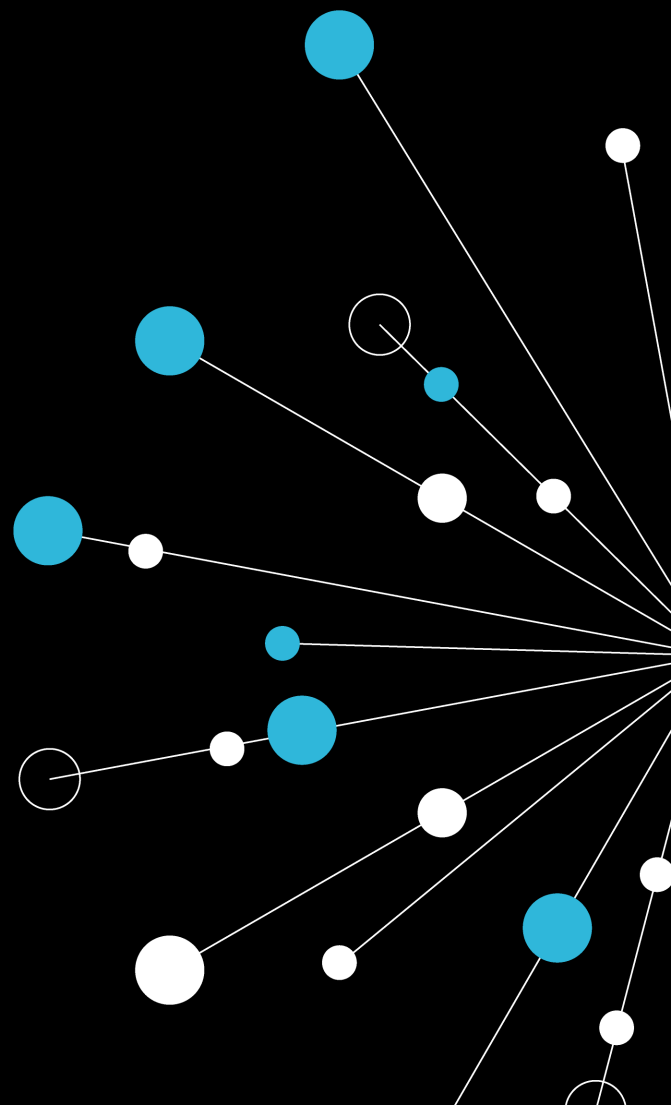


Implementing the Solution

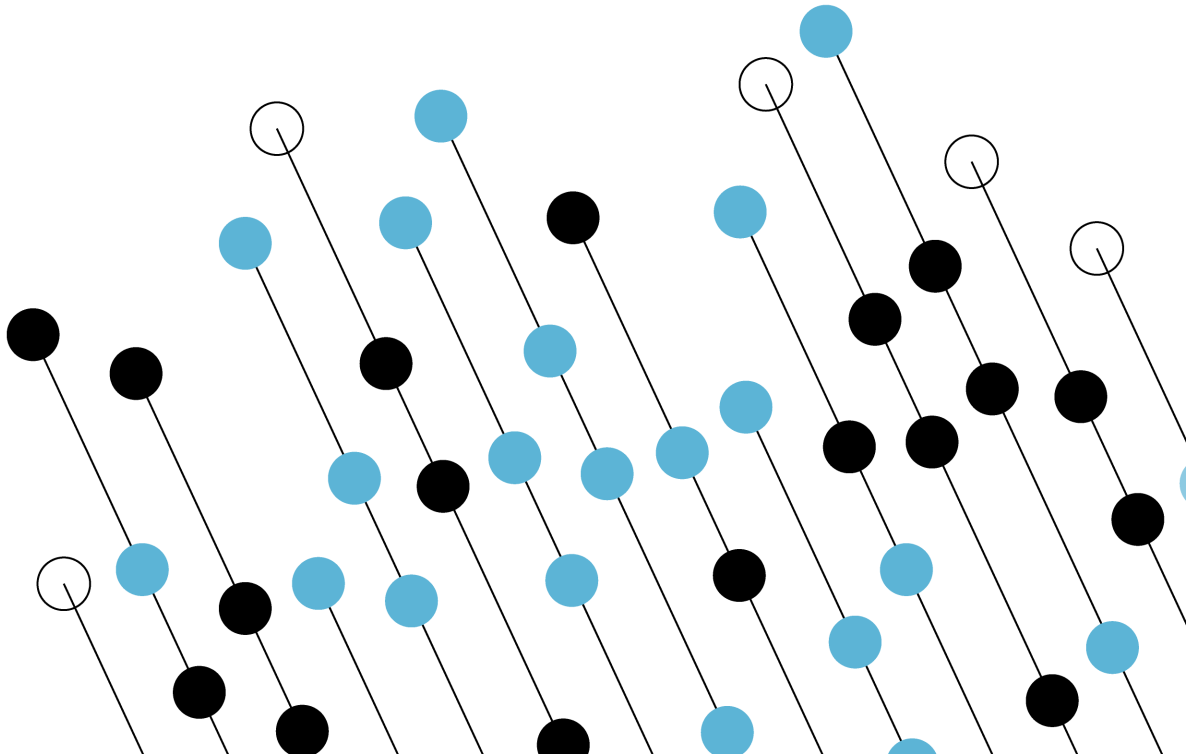
Activity 9

What are the key points you need to consider when implementing a solution?

In your Learner Activity Document, screenshot your diagram from Miro and write up a short summary.



Documentation and Knowledge Sharing



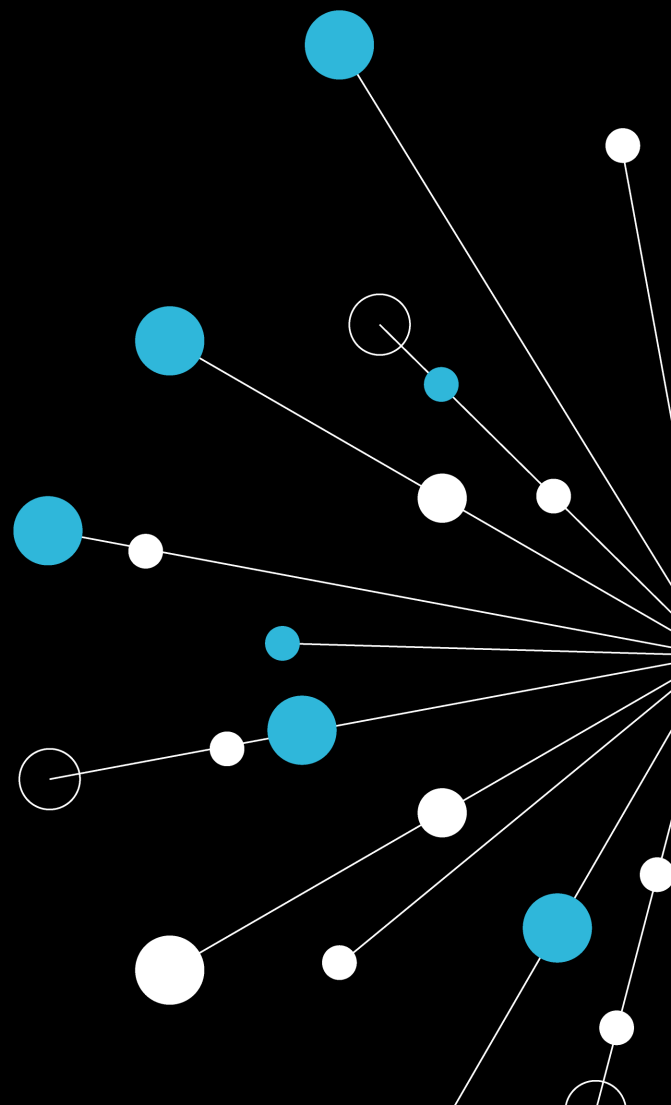
Documentation and Knowledge Sharing

Activity 10

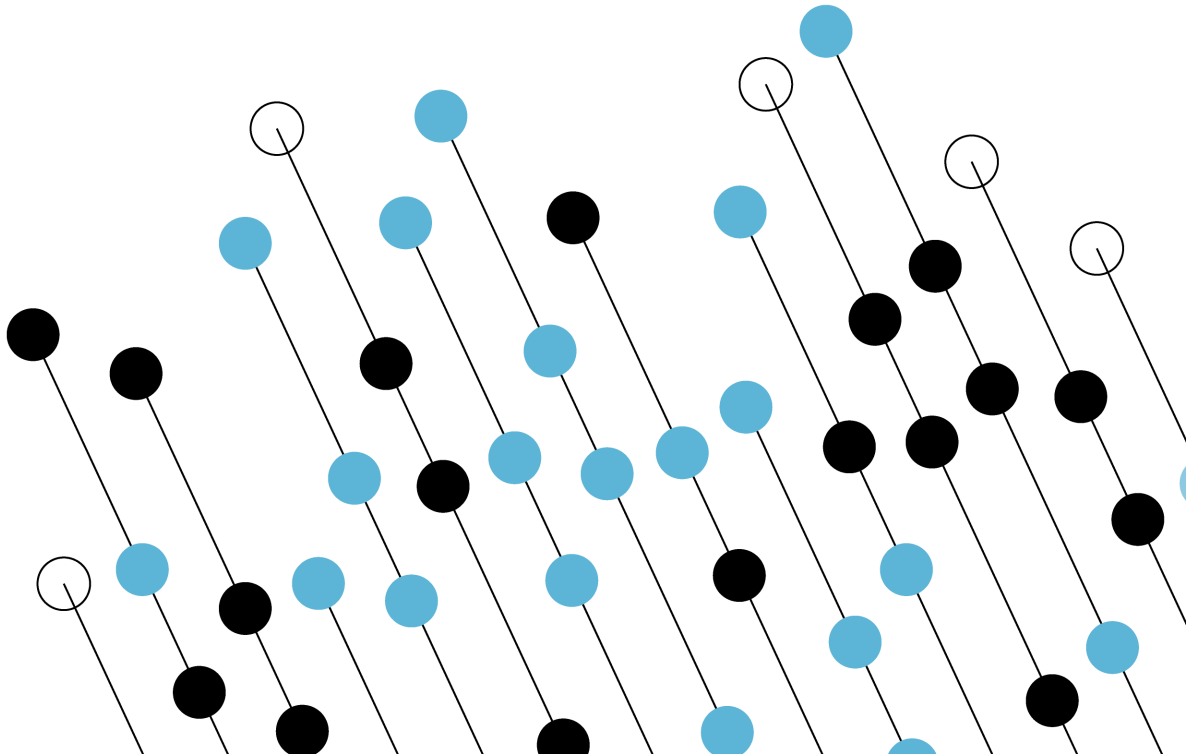
What are the key points you need to consider when creating follow up documentation?

How does your current organisation share knowledge? What are the key benefits of knowledge sharing?

In your Learner Activity Document, screenshot your diagram from Miro and write up a short summary.



Problem-Solving Reflection



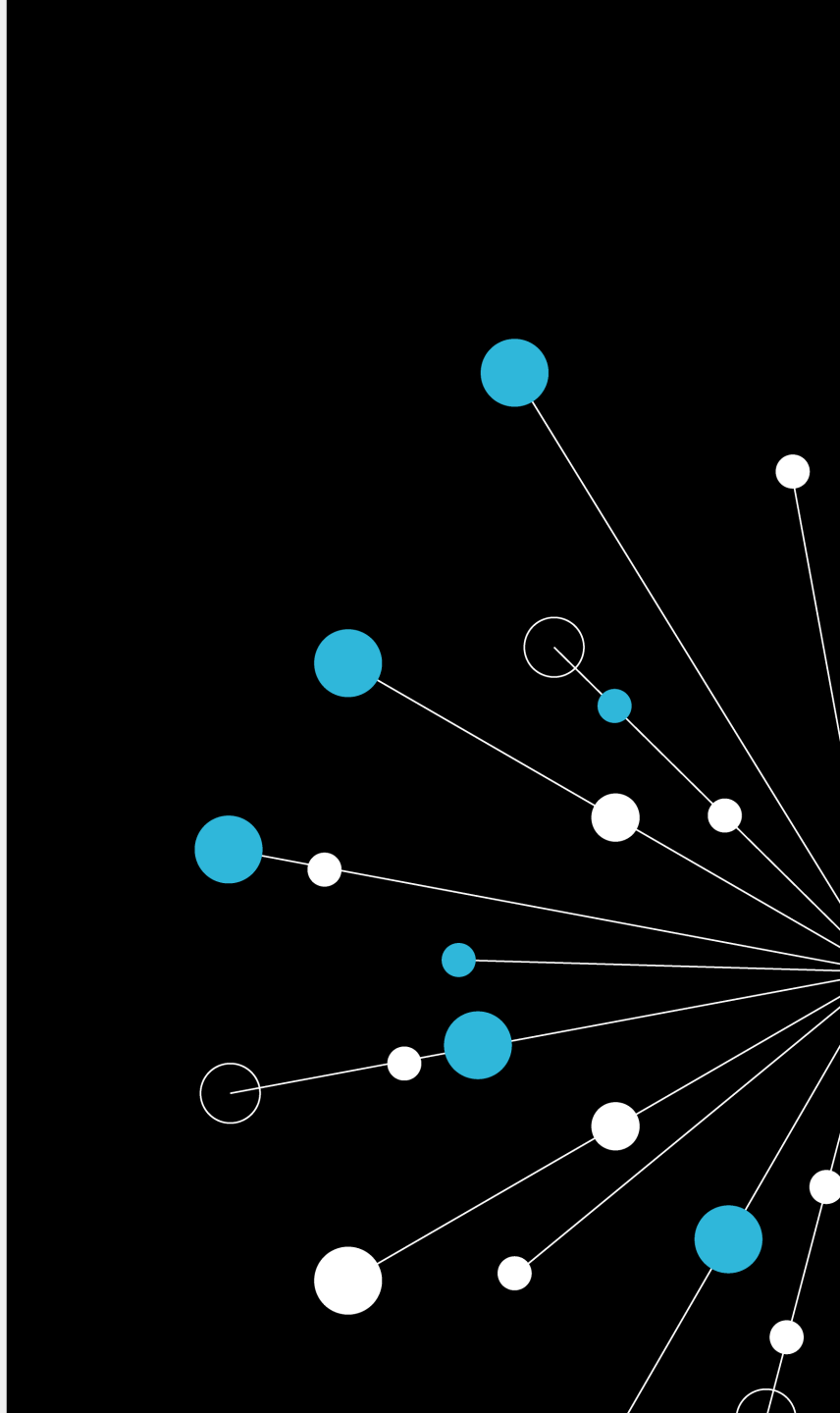
Problem-Solving Reflection

Activity 11

How are you going to use problem solving in the future:

- Continue doing?
- Start doing?
- Stop doing?

Write up your thoughts/reflections on what you have worked through today and when completed, upload to Bud.



End of Module Review

- Kindly upload your completed Delegate Guide and *Certification of Completion to Bud under the following module:

- Activity 2.3 | Discover and Practise | Core IT Principles: Legislation and IT Systems (Live Event)

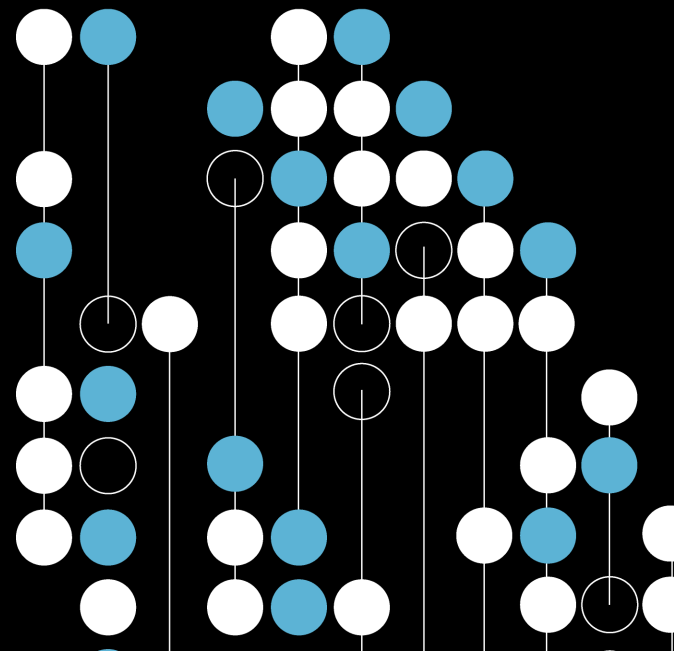
*Certification of Completion can be obtained by going to the live event module in Bud and pressing “Start now”, then ticking off the modules accordingly.

Learning instructions & materials

To complete this activity:

1. Attend a two-day live event (*), led by your tutor | Start now | - **please don't start this learning path before the live event.**
2. Submit your work in the 'Messages and Submissions' area above

QA

End of Module Review

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